

Bayes-Error-Driven Iterative Supervised Feature Selection

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Abstract—Supervised feature selection is essential for improving model interpretability and predictive performance, particularly in high-dimensional settings where irrelevant and redundant features can degrade classification accuracy. In this paper, we propose Bayes Error Driven Iterative Supervised Feature Selection (B-ISFS), a novel algorithm that iteratively selects features by leveraging the Gram-Schmidt orthogonalization process in function spaces. B-ISFS constructs an orthonormal basis of functions such that each selected feature maximally contributes to the classification task via the Bayes predictor. At each iteration, the algorithm updates the feature space by removing the projections of the remaining features onto the previously selected orthonormal functions, thereby identifying the feature that minimizes the Bayes misclassification error. By operating in function space, B-ISFS effectively captures nonlinear dependencies among features. We evaluate the proposed method on real-world datasets and demonstrate its superiority in classification accuracy compared to state-of-the-art feature selection techniques.

Index Terms—Supervised Feature Selection, Gram-Schmidt Orthogonalization, Bayes Classifier, Bayes Error Rate.

I. INTRODUCTION

SUPERVISED feature selection is a critical step in machine learning and data analysis. It uses labeled data to identify features that best predict the target outcome while eliminating redundant or irrelevant features [1]–[3]. By focusing on the most informative features, these methods enhance generalization, reduce overfitting and computational cost, particularly in high-dimensional settings where features outnumber samples [4]–[7]. They also improve interpretability [8]–[10], lower data acquisition and storage costs, and help mitigate the curse of dimensionality. These benefits make feature selection indispensable in finance [11], healthcare [12], and genomics [13].

Supervised feature selection spans several methodological families. Similarity-based methods, such as Relief algorithms, rank features based on how well nearest-neighbor relationships align with class labels [14], [15]. Kernel methods capture nonlinear structure by projecting data into high-dimensional feature spaces [16]. Although powerful, deep learning based feature selection methods are computationally expensive and highly sensitive to hyperparameter tuning. [17], [18]. Information-theoretic criteria, including mRMR, mutual information estimators, and the maximum information coefficient, quantify feature-label dependence using entropy [19]–[21]. Supervised Fourier Feature Selection uses Fourier parity functions to eliminate nonlinear redundancy and select informative features [22], [23].

Feature selection methods are typically categorized as filters, wrappers, or embedded techniques [24]. Filter methods

rank features using model-agnostic statistics such as correlation, mutual information, chi-square, or Fourier criteria [19], [22], [25]–[27]. Wrapper methods evaluate feature subsets with the learning algorithm itself, as in recursive feature elimination or forward-backward search [28]–[31], often improving accuracy but at higher computational cost and overfitting risk. Embedded methods incorporate selection into model training via sparsity penalties or tree-based pruning [32]–[34], balancing efficiency and predictive performance.

In this paper, we address the problem of supervised feature selection by developing a principled algorithm that effectively captures nonlinear dependencies in the data. Specifically, given a dataset consisting of independent and identically distributed samples from a random variable $X \in \mathbb{R}^d$ with binary labels $Y \in \{-1, 1\}$, our objective is to select a subset of features that maximizes information about Y while eliminating nonlinear redundancies among features. To this end, we propose the *Bayes Error Driven Iterative Supervised Feature Selection (B-ISFS)* algorithm, which is based on the Gram-Schmidt (GS) orthogonalization process in function spaces, with functions in $L^2(P_X)$, the space of square-integrable functions with respect to the marginal distribution of X . B-ISFS begins by selecting the feature that minimizes the Bayes misclassification error when used individually. It then applies Gram-Schmidt orthogonalization to each remaining feature with respect to the span of the functions associated with all previously chosen features, thereby removing nonlinear redundancies in the data. At each iteration, the feature whose orthogonalized representation achieves the lowest Bayes error rate is added to the selected set, and this process is repeated until the desired number of features is obtained. The mathematical justification for using orthogonalized functions lies in their role in computing the Bayes misclassification error conditioned on previously chosen features, ensuring an effective and principled selection procedure. B-ISFS is primarily inspired by our recent Gram-Schmidt methods for unsupervised feature extraction and selection [35]–[37]. This work can be viewed as a supervised extension of the feature selection algorithm introduced in [35], [37], extending it to the supervised setting by incorporating Bayes error minimization as the guiding criterion.

The remainder of this paper is organized as follows. Section II introduces the definitions and notations. Section III reviews the Gram-Schmidt orthogonalization process in function spaces. Section IV presents the proposed feature selection algorithm. Section V evaluates the performance of the proposed algorithm on benchmark datasets and compares it with the state-of-the-art supervised feature selection algorithms.

II. PRELIMINARY DEFINITIONS AND NOTATIONS

Let $X = [X_1, \dots, X_d]^T \in \mathbb{R}^d$ be a zero mean random vector, and let $Y \in \{-1, 1\}$ denote its binary label. Samples

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(X, Y) are assumed to be independent and identically distributed. Deterministic vectors are written in bold lowercase, e.g., $\mathbf{x} = [x_1, \dots, x_d]^\top$. For any subset $\mathcal{S} \subseteq [d] = \{1, \dots, d\}$, we denote by $X_{\mathcal{S}}$ the sub-vector of X consisting of coordinates indexed by $\mathcal{S}$. We work with a function family $\mathcal{F}(\mathbf{x})$ whose elements depend on the coordinates of $\mathbf{x}$ and include the singleton functions $\{f_i(\mathbf{x}) = x_i\}_{i=1}^d$. For any subset $\mathcal{S} \subseteq [d]$, let $\mathcal{F}(\mathbf{x}_{\mathcal{S}})$ denote the functions that depends only on $\{x_i : i \in \mathcal{S}\}$. We assume throughout that $\mathcal{F}(X_{\mathcal{S}}) \subseteq L^2(P_X)$, the space of square-integrable functions with respect to the marginal distribution of X . B-ISFS operates in a Hilbert space generated by $\mathcal{F}$. For any feature subset $\mathcal{S}$, we denote by $\hat{\mathcal{F}}(X_{\mathcal{S}})$ the orthogonalized functions in $\mathcal{F}(X_{\mathcal{S}})$, which preserves the linear span. We use the assignment notation $a \leftarrow b$ to indicate that the value b is computed and stored in a . At iteration j , let $\bar{\mathcal{S}} = \{\bar{s}_1, \dots, \bar{s}_{d-j}\}$ denote the set of remaining (unselected) features. For each $\bar{s}_i$, we construct an orthonormal set $\hat{\mathcal{H}}_{\bar{s}_i}^{(j+1)}$ by orthogonalizing $\mathcal{F}(X_{\bar{s}_i})$ against $\hat{\mathcal{F}}(X_{\mathcal{S}})$. Collecting these sets yields $\hat{\mathcal{H}}^{(j+1)} = \{\hat{\mathcal{H}}_{\bar{s}_i}^{(j+1)}\}_{i=1}^{d-j}$, whose elements are orthogonal to the current span, $\hat{\mathcal{F}}(X_{\mathcal{S}})$.

Remark 1. *In this paper, we work with random variables such as X and Y without assuming explicit knowledge of their underlying distributions. These variables are treated symbolically and are used primarily to define and compute expectations. In practice, all expectations are approximated using empirical means.*

III. GRAM-SCHMIDT ORTHOGONALIZATION

In this section, we describe the Gram-Schmidt (GS) orthogonalization process over function spaces, which forms the computational core of our feature selection method. In the context of our feature selection algorithm, the Gram-Schmidt process described in Algorithm 1 takes as input an orthogonalized function family $\hat{\mathcal{F}}(X_{\mathcal{S}})$ constructed from the previously selected features $X_{\mathcal{S}}$. It also receives a set of candidate functions $\mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]})$ drawn from the predefined function family $\mathcal{F}(\mathbf{x})$, where $\mathbf{x} = [x_1, \dots, x_m]^\top$ is a vector of non-random variables and m is the number of selected features. These candidate functions depend on one additional variable x_{j+1} and are denoted by $\{f_1(\mathbf{x}), \dots, f_n(\mathbf{x})\}$, where $f_1(\mathbf{x}) = x_{j+1}$ always exists in the function sets considered in the orthogonalization process (Line 2). The input further includes the index set $\mathcal{S} = \{s_1, \dots, s_j\}$ corresponding to the currently selected features, and a placeholder variable Z , which will be substituted with each candidate feature $X_{\bar{s}_\ell}$ for $\ell = 1, \dots, d-j$ during the orthogonalization step.

For each function in $\mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]})$, the algorithm constructs a new function g by substituting $x_i \leftarrow X_{s_i}$ for all $i \in [j]$ and $x_{j+1} \leftarrow Z$, where Z serves as a placeholder for each unselected feature $X_{\bar{s}_\ell}$ (Line 4). The function g is then orthogonalized by subtracting its projection onto the span of the current basis $\mathcal{T}$ (Line 5), where the projection coefficients are computed as inner products $\mathbb{E}[gf]$ for each $f \in \mathcal{T}$. The resulting function is normalized to unit norm (Line 6), ensuring that the output basis remains orthonormal. The normalized function is then added to the orthogonalized set $\mathcal{T}$ (Line 7). Once all functions in the candidate set have

been processed through this procedure, the algorithm returns the updated orthogonalized function family $\mathcal{T}$ (Line 9).

Algorithm 1 $\text{GS}(\hat{\mathcal{F}}(X_{\mathcal{S}}), \mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]}), \mathcal{S})$

Require:

- $\hat{\mathcal{F}}(X_{\mathcal{S}})$: an orthogonalized function family in the random variables $X_{\mathcal{S}}$.
- $\mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]})$: the set of functions in $\mathcal{F}(\mathbf{x})$ that depend on $\mathbf{x}_{[j+1]}$ but not solely on $\mathbf{x}_{[j]}$.
- $\mathcal{S}$: the indices of variables $X_{\mathcal{S}}$ that appear in $\hat{\mathcal{F}}(X_{\mathcal{S}})$.
- Z : a new random variable.

Ensure: Orthogonalized function family $\mathcal{T}$.

- 1: **Initialize:** $\mathcal{T} \leftarrow \hat{\mathcal{F}}(X_{\mathcal{S}})$.
 - 2: Denote $\mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]}) \triangleq \{f_1(\mathbf{x}), \dots, f_n(\mathbf{x})\}$, where $f_1(\mathbf{x}) = x_{j+1}$.
 - 3: **for** $\ell \leftarrow 1$ **to** n **do**
 - 4: $g(X_{\mathcal{S}}, Z) \leftarrow f_\ell(X_{\mathcal{S}}, Z)$ (Create g from f_ℓ by substituting $x_i \leftarrow X_{s_i}$ for $i \in [j]$ and $x_{j+1} \leftarrow Z$.)
 - 5: $g(X_{\mathcal{S}}, Z) \leftarrow g(X_{\mathcal{S}}, Z) - \sum_{f \in \mathcal{T}} \mathbb{E}[gf]f$ (Orthogonalize g with respect to $\mathcal{T}$.)
 - 6: $g(X_{\mathcal{S}}, Z) \leftarrow \frac{g(X_{\mathcal{S}}, Z)}{\|g(X_{\mathcal{S}}, Z)\|}$ (Normalize g .)
 - 7: $\mathcal{T} \leftarrow \mathcal{T} \cup \{g(X_{\mathcal{S}}, Z)\}$ (Add g to $\mathcal{T}$.)
 - 8: **end for**
 - 9: **Return** $\mathcal{T}$.
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IV. BAYES ERROR DRIVEN ITERATIVE SUPERVISED FEATURE SELECTION

In this section, we present the Bayes Error Driven Iterative Supervised Feature Selection (B-ISFS) algorithm, outlined in Algorithm 2, which selects features in an iterative manner based on Gram-Schmidt orthogonalization. Given a dataset (X, Y) with standardized features—i.e., each feature is centered by subtracting its mean and then scaled by dividing by its standard deviation, yielding variables with zero mean and unit variance—a function family $\mathcal{F}(\mathbf{z})$, and a desired number of features m , the algorithm returns a subset $\mathcal{S} = \{s_1, \dots, s_m\}$ of selected feature indices.

The algorithm begins by computing an initial score parameter $\rho_i^{(1)} = \mathbb{E}[\mathbb{E}[YX_i]X_i]$ for all features, i.e., $i \in [d]$, which quantifies the predictive strength of feature X_i when used in isolation (Line 1). This step corresponds to selecting the feature that minimizes the Bayes misclassification error when using a univariate predictor. The orthogonalized function families are initialized as $H_i^{(1)} = \{X_i\}$ for all $i \in [d]$, and the selected set $\mathcal{S}$ is initially empty.

For each iteration $j = 1, \dots, m$ (Line 2), the algorithm selects the index s_j of the feature with the highest score among those not yet selected, i.e., $s_j = \arg \max_{i \in [d] \setminus \mathcal{S}} \rho_i^{(j)}$ (Line 3), and adds it to the selected set $\mathcal{S}$ (Line 4). The set of remaining feature indices is then updated as $\bar{\mathcal{S}} = [d] \setminus \mathcal{S} = \{\bar{s}_1, \dots, \bar{s}_{d-j}\}$ (Line 5). Next, the orthogonal basis from the previously selected feature s_j is retrieved and denoted $\hat{\mathcal{F}}(X_{\mathcal{S}}) = \hat{\mathcal{H}}_{s_j}^{(j)}$ (Line 6). For each remaining feature $\bar{s}_\ell \in \bar{\mathcal{S}}$, the algorithm applies the Gram-Schmidt procedure to compute an orthonormal function set $\hat{\mathcal{H}}_{\bar{s}_\ell}^{(j+1)}$ by orthogonalizing the function family $\mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]})$ against the previously

orthogonalized functions (Line 8). This step is performed independently for each candidate feature and can be parallelized (Line 7). Once the orthogonal basis $\hat{\mathcal{H}}_{\bar{s}_\ell}^{(j+1)}$ is constructed for each candidate feature, the algorithm computes a new score $\rho_{\bar{s}_\ell}^{(j+1)}$ by projecting the label Y onto the new orthogonalized functions, excluding those already in the selected basis. The score is given by $\rho_{\bar{s}_\ell}^{(j+1)} = \mathbb{E}[\|\sum_{f \in \hat{\mathcal{H}}_{\bar{s}_\ell}^{(j+1)} \setminus \hat{\mathcal{F}}(X_S)} \mathbb{E}[Yf]f\|]$, which quantifies the contribution of the candidate feature $\bar{s}_\ell$ to predicting Y after removing redundancy with previously selected features (Line 9). The algorithm repeats this process for m iterations, each time selecting the feature that maximizes predictive gain under orthogonalized representation. B-ISFS thus follows a greedy strategy to minimize Bayes misclassification error, incrementally constructing a set of non-redundant, highly informative features.

Algorithm 2 B-ISFS

Require: Dataset (X, Y) with standardized features, a function family $\mathcal{F}(\mathbf{z})$, and the desired number of selected features m .

Ensure: The indices of selected features $\mathcal{S} = \{s_1, \dots, s_m\}$.

- 1: **Initialize:** Compute $\rho_i^{(1)} = \mathbb{E}[\|\mathbb{E}[YX_i]X_i\|]$ and $H_i^{(1)} = \{X_i\}$ for all $i \in [d]$ and $\mathcal{S} = \emptyset$.
 - 2: **for** $j \leftarrow 1$ **to** m **do**
 - 3: Let $s_j \triangleq \arg \max_{i \in [d] \setminus \mathcal{S}} \rho_i^{(j)}$.
 - 4: Add s_j to $\mathcal{S}$.
 - 5: Let $\bar{\mathcal{S}} = [d] \setminus \mathcal{S} = \{\bar{s}_1, \dots, \bar{s}_{d-j}\}$.
 - 6: Set $\hat{\mathcal{F}}(X_S) = \hat{\mathcal{H}}_{s_j}^{(j)}$.
 - 7: **for each** $\ell \in \{1, \dots, d-j\}$ **(in parallel)** **do**
 - 8: $\hat{\mathcal{H}}_{\bar{s}_\ell}^{(j+1)} \leftarrow \text{GS}(\hat{\mathcal{F}}(X_S), \mathcal{F}(\mathbf{x}_{[j+1]}) \setminus \mathcal{F}(\mathbf{x}_{[j]}), \mathcal{S})$.
 - 9: $\rho_{\bar{s}_\ell}^{(j+1)} \leftarrow \mathbb{E}[\|\sum_{f \in \hat{\mathcal{H}}_{\bar{s}_\ell}^{(j+1)} \setminus \hat{\mathcal{F}}(X_S)} \mathbb{E}[Yf]f\|]$.
 - 10: **end for**
 - 11: **end for**
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The following theorem establishes the theoretical foundation for Algorithm 2 and presents the central result of this work.

Theorem 1. Let $X \in \mathbb{R}^d$ be the data random variable and let $Y \in \{-1, 1\}$ be the target random variable (label). Let $\mathcal{S}$ denote the subset of features selected by Algorithm 2, with $|\mathcal{S}| = m$. Let $\hat{\mathcal{F}}(X_S)$ denote the set of orthonormal functions obtained by applying Algorithm 2 to the function family $\mathcal{F}(\mathbf{z})$. Then, the minimum misclassification probability achievable using the m selected features is given by

$$\Pr_{\text{mis}} = \frac{1}{2} \left(1 - \max_{\substack{\mathcal{S} \subseteq [d] \\ |\mathcal{S}|=m}} \mathbb{E} \left[\left\| \sum_{f \in \hat{\mathcal{F}}(X_S)} c_f f \right\| \right] \right), \quad (1)$$

where $c_f = \mathbb{E}[Yf]$ for each $f \in \hat{\mathcal{F}}(X_S)$.

The key to proving Theorem 1 lies in a lemma that characterizes the relationship between the Gram-Schmidt orthogonalization of the function family $\mathcal{F}(\mathbf{z})$ and the Bayes optimal predictor using m selected features. We now state the lemma that serves as the foundation for the proof of Theorem 1.

Lemma 1. Let $X \in \mathbb{R}^d$ be the data random variable, let $Y \in \{-1, 1\}$ be the target random variable (label), and let $\mathcal{S}$ be the index set of features selected by Algorithm 2

with $|\mathcal{S}| = m$. Denote by $\hat{\mathcal{F}}(X_S) \subseteq L^2(\sigma(X_S))$ the set of orthogonalized functions obtained by applying Algorithm 2 to the function family $\mathcal{F}(\mathbf{z})$. Assume that the function family $\mathcal{F}(X_S)$ is sufficiently rich (dense in the L^2 -sense) such that the regression (conditional expectation) function $\eta(X_S) = \mathbb{E}[Y|X_S]$ can be approximated well (in L^2 -norm) by linear combinations of orthogonalized functions of $\mathcal{F}(X_S)$. Then, for any observation $X_S = \alpha$, the Bayes classifier that uses only X_S is given by $h^*(\alpha) = \text{Sign} \left(\sum_{f(X_S) \in \hat{\mathcal{F}}(\alpha)} c_f f(\alpha) \right)$, where $c_f = \mathbb{E}[Yf(X_S)]$ for each $f(X_S) \in \hat{\mathcal{F}}(X_S)$.

Proof of Lemma 1. The Bayes classifier is the decision rule that minimizes the probability of misclassification. In binary classification, with $Y \in \{-1, 1\}$, the Bayes classifier using only features X_S for a given observation $X_S = \alpha$ assigns

$$h^*(\alpha) = \arg \max_{y \in \{-1, 1\}} \Pr(Y = y | X_S = \alpha).$$

For a fixed observation $X_S = \alpha$, we have

$$\begin{aligned} \eta(X_S = \alpha) &= \mathbb{E}[Y | X_S = \alpha] \\ &= \Pr(Y = 1 | X_S = \alpha) - \Pr(Y = -1 | X_S = \alpha) \\ &= 2 \Pr(Y = 1 | X_S = \alpha) - 1. \end{aligned}$$

Hence $\Pr(Y = 1 | X_S = \alpha) > 0.5$ if and only if $\eta(X_S = \alpha) > 0$. Therefore,

$$h^*(\alpha) = \text{Sign}(\eta(X_S = \alpha)) = \text{Sign}(\mathbb{E}[Y | X_S = \alpha]).$$

By hypothesis, $\mathcal{F}(X_S)$ is dense (in L^2) enough that $\eta(X_S)$ can be approximated well by linear combinations of orthonormalized functions in $\hat{\mathcal{F}}(X_S)$. Consequently, $\eta(X_S)$ admits an expansion $\eta(X_S = \alpha) = \sum_{f(X_S) \in \hat{\mathcal{F}}(X_S)} c_f f(\alpha)$. The coefficient corresponding to each $f(X_S) \in \hat{\mathcal{F}}(X_S)$ is

$$\begin{aligned} c_f &= \mathbb{E}[\eta(X_S)f(X_S)] = \mathbb{E}[\mathbb{E}[Y|X_S]f(X_S)] \\ &\stackrel{(\star)}{=} \mathbb{E}[\mathbb{E}[Yf(X_S)|X_S]] \\ &\stackrel{(\star\star)}{=} \mathbb{E}[Yf(X_S)], \end{aligned}$$

where $(\star)$ holds because $f(X_S)$ is $\sigma(X_S)$ -measurable, and $(\star\star)$ is the law of total expectation. Substituting these c_f into the expansion of η yields

$$\eta(X_S = \alpha) = \sum_{f(X_S) \in \hat{\mathcal{F}}(X_S)} \mathbb{E}[Yf(X_S)]f(\alpha).$$

Finally, since $h^*(\alpha) = \text{Sign}(\eta(X_S = \alpha))$, we conclude

$$h^*(\alpha) = \text{Sign} \left(\sum_{f(X_S) \in \hat{\mathcal{F}}(X_S)} \mathbb{E}[Yf(X_S)]f(\alpha) \right),$$

as required. $\square$

Proof of Theorem 1. For any fixed subset $\mathcal{S} \subseteq [d]$ of size m , let $h : X_S \rightarrow \{-1, 1\}$ be an arbitrary classifier whose input is the features in set $\mathcal{S}$ and whose output is binary. Since $Y \in \{-1, 1\}$, the misclassification probability for this classifier can be expressed as $\Pr\{Y \neq h(X_S)\} = \frac{1}{2} - \frac{1}{2} \mathbb{E}[Yh(X_S)]$. Thus minimizing the misclassification probability is equivalent to maximizing the correlation $\mathbb{E}[Yh(X_S)]$. From Lemma 1, recall that the Bayes predictor for the features in set $\mathcal{S}$, which minimizes the misclassification probability, is given by $h^*(X_S = \alpha) = \text{Sign}(\eta(\alpha))$ for all $\alpha \in \mathcal{X}_S$, where $\mathcal{X}_S$

denotes the sample space of the random variable X_S and $\eta(X_S) = \mathbb{E}[Y|X_S]$. Substituting h^* into the above expression yields

$$\Pr\{Y \neq h^*(X_S)\} = \frac{1}{2} - \frac{1}{2} \mathbb{E}[\text{Sign}(\eta(X_S))].$$

Conditioning on X_S and using the fact that $\text{Sign}(\eta(X_S))$ is measurable with respect to X_S , we obtain

$$\begin{aligned} \mathbb{E}[\text{Sign}(\eta(X_S))] &= \mathbb{E}[\text{Sign}(\eta(X_S)) \mathbb{E}[Y | X_S]] \\ &= \mathbb{E}[\text{Sign}(\eta(X_S)) \eta(X_S)] \\ &= \mathbb{E}[|\eta(X_S)|]. \end{aligned}$$

Therefore, we obtain, it follows that

$$\Pr\{Y \neq h^*(X_S)\} = \frac{1}{2} - \frac{1}{2} \mathbb{E}[|\eta(X_S)|].$$

Lemma 1 further guarantees the Gram-Schmidt orthonormal expansion for $\eta(X_S)$. Hence

$$\mathbb{E}[|\eta(X_S)|] = \mathbb{E}\left[\left|\sum_{f \in \hat{\mathcal{F}}(X_S)} c_f f\right|\right].$$

Finally, because minimizing $\Pr\{Y \neq h^*(X_S)\}$ over all subsets of size m is equivalent to maximizing $\mathbb{E}[|\eta(X_S)|]$, we conclude

$$\begin{aligned} \Pr_{\text{mis}} &= \min_{S \subseteq [d]} \Pr\{Y \neq h^*(X_S)\} \\ &= \frac{1}{2} \left(1 - \max_{\substack{S \subseteq [d] \\ |S|=m}} \mathbb{E}\left[\left|\sum_{f \in \hat{\mathcal{F}}(X_S)} c_f f\right|\right]\right), \end{aligned}$$

which completes the proof. $\square$

The maximization problem in (1) is a combinatorial optimization problem, which is computationally intractable in general. To address this challenge, we propose a greedy iterative algorithm. In the first iteration, the algorithm selects the feature that maximizes $\mathbb{E}[|\eta(X_S)|]$ as defined in (1). Subsequently, all remaining features and their associated functions in $\mathcal{F}(\mathbf{z})$ are orthogonalized with respect to the selected feature. Among these orthogonalized functions, the algorithm selects the feature that again maximizes $\mathbb{E}[|\eta(X_S)|]$. This procedure is repeated until the desired number of features is selected.

V. EXPERIMENTAL RESULTS

In this section, the theoretical results and performance of B-ISFS are evaluated through simulation studies on real-world datasets, where expectations are approximated using empirical means. We validate the performance of B-ISFS for classification tasks on benchmark datasets taken from the UCI repository [38] and [39]. The properties of the benchmark datasets used in the experiments are provided in Table I. In Figure 1, we evaluate the performance of B-ISFS

TABLE I
PROPERTIES OF THE TESTED BENCHMARK DATASETS.

Dataset	USPS	COIL-20	Isolet
Features	256	1024	617
Samples	7291	1440	1559

using multilinear polynomials up to degree 3, comparing it against several well-known feature selection algorithms: Conditional Mutual Information Maximization (CMIM) [40], Fast

Correlation-Based Filter (FCBF) [41], minimal-redundancy-maximal-relevance (mRMR) [19], ReliefF [42], and Recursive Feature Selection (RFS) [43]. We show the classification accuracy for different numbers of selected features, ranging from 5 to 100, with steps of 5 features on benchmark datasets demonstrate that B-ISFS outperforms these other algorithms, showcasing its superior ability to select informative features for improved predictive performance.

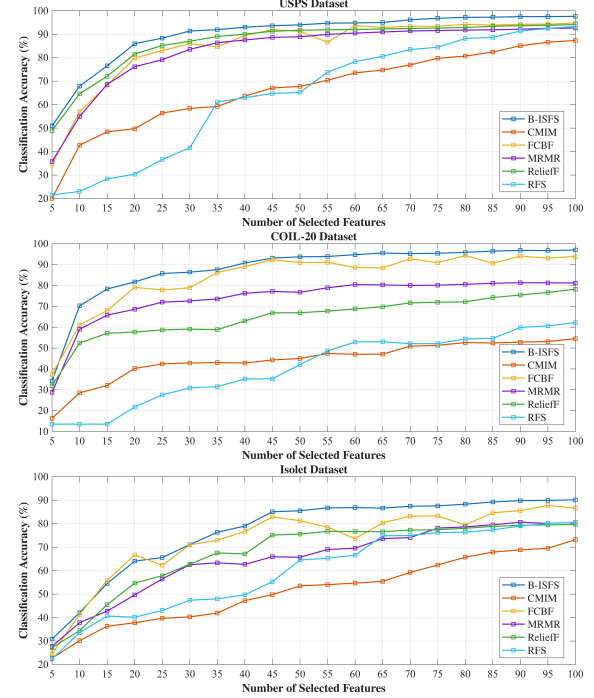


Fig. 1. Classification accuracy (%) of B-ISFS with multilinear polynomials up to degree 3 compared to CMIM, FCBF, mRMR, ReliefF, and RFS on the benchmark datasets.

We employed a support vector machine classifier with a radial basis function kernel for the classification tasks. To evaluate the performance of the algorithms, we applied 5-fold cross-validation on the entire datasets, ensuring robust validation and minimizing overfitting. For multi-class classification tasks, we utilized the One-vs-Rest strategy, where a binary classifier is trained for each class against all other classes, enabling the model to handle multiple classes effectively.

VI. CONCLUSION

In this paper, we introduced Bayes Error Driven Iterative Supervised Feature Selection (B-ISFS), a novel algorithm that employs Gram-Schmidt orthogonalization in function spaces to iteratively select the most informative features for classification tasks. By constructing an orthonormal function basis and minimizing the Bayes misclassification error at each iteration, B-ISFS effectively captures nonlinear dependencies among features while eliminating redundancy. The theoretical foundation of the algorithm is rigorously established, demonstrating its connection to the optimal Bayes predictor and its ability to enhance class separability by selecting features that optimize classification performance. Experimental results on benchmark datasets further validate the superiority of B-ISFS over several well-known feature selection methods.

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